Chapter 11

Banking risks

Learning objectives

- To define the most common risks in banking
- To distinguish between the various risks in banking
- To appreciate the relevance and urgency of sustainability risks
- To understand the importance of the interrelation among banking risks

11.1 Introduction

For any privately owned bank, management's goal is to maximise shareholders' value while ensuring that the bank is safe and sound. If the institution is publicly listed and markets are efficient, returns are proportional to the risks taken; if the bank is small and unlisted, managers will try to maximise the value of the owner's investments by seeking the highest returns for what they deem to be acceptable levels of risk. With increased pressure on banks to improve shareholders' returns, banks have had to assume higher risks and, at the same time, manage these risks to avoid losses. The processes of deregulation, globalisation and conglomeration have offered productive opportunities for banks in terms of profitability and value creation activities (see Chapter 9) but have also posed serious risk challenges. The financial crisis in the second half of the 2000s has demonstrated on one hand the fragilities of the banking sector globally and, on the other, the potential systemic risks associated with the interconnectedness of banking activities. In this chapter we describe the main types of risks modern banks have to face. We also discuss why bank managers are under increased pressure to identify the conditions that are linked directly or indirectly to sustainability issues and that pose risks that can materialise in the long term, such as climate change. An introduction to the prevailing risk management techniques will be provided in more detail in Chapter 12.

11.2 Credit risk

According to the Basel Committee on Banking Supervision (2000) **credit risk** is defined as 'the potential that a bank borrower or counterparty will fail to meet its obligations in accordance with agreed terms'. Generally credit risk is associated with the traditional lending activity of banks and it is simply described as the risk of a loan not being repaid in part or in full. However, credit risk can also derive from holding bonds and other securities.

Credit risk is the risk of a decline in the credit-standing of a counterparty. Such deterioration does not imply default, but means that the probability of default increases. Capital markets value the credit-standing of firms through the rate of interest charged on bonds or other debt issues, changes in the value of shares, and ratings provided by the **credit-rating agencies** (such as Standard & Poor's, Moody's and Fitch). However, banks also face credit risk in a number of other financial instruments such as derivative products and guarantees. This particular type of credit risk is sometimes referred to as **counterparty risk**. Resti and Sironi (2007) define the risk associated with deterioration in the counterparty's creditworthiness as a 'migration' risk that becomes effectively a 'downgrading' risk if the rating agency that issued the public credit rating downgraded it. According to the authors, in the context of credit risk, it is also useful to distinguish between spread risk, recovery risk, pre-settlement (or substitution risk) and country risk (Section 11.7). Spread risk is the risk connected with an increase in the spreads required of borrowers, such as bond issuers, by the market. When risk aversion increases due for example to instability, crisis or shocks, the spread differential between good quality and bad quality bonds may rise (so-called 'flight to quality'). In the case of insolvent assets of the counterparty, there is also the recovery risk that can occur when the recovery rate is lower than initially predicted because of lower liquidation value or unexpected delays in cashing in bad assets. A bank also runs a pre-settlement or substitution risk, which is the risk that may occur when the bank counterparty in an over the counter derivative transaction becomes insolvent prior to maturity, forcing the bank to substitute it at less favourable conditions.

In general terms, bank managers should minimise credit losses by building a portfolio of assets (loans and securities) that diversifies the degree of risk. This is because very low default risk assets are associated with low credit risk and low expected return, while higher expected return assets have a higher probability of default (i.e. a higher credit risk). Focusing on loans that constitute the largest proportion of a bank's assets, in Chapter 10 we mentioned that the traditional lending function involves four different components (Sinkey, 2002; Greenbaum *et al*., 2019):

1. originating (the application process);
2. funding (approving the loan and availing funds);
3. servicing (collecting interest and principal payments);
4. monitoring (checking on borrowers' behaviour through the life of the loan).

Banks must investigate borrowers' ability to repay their loans before and after the loan has been made (these activities are known as screening and monitoring functions of banks) because of their aim to maximise value and the responsibility they have towards their depositors and deposit insurers to be safe and sound.

To recall, banks are said to act as *delegated monitors* on behalf of lenders and to this end they use technology and innovation in the design and enforcement of contracts. These contracts are costly for the banks but essential for the protection of both banks' owners and lenders. However, *agency problems* may arise as a result of functions (1) and (4). As shown in Chapter 1, agency problems imply potential contractual frictions between principals (lenders) and agents (borrowers) because of asymmetric information, moral hazard and adverse selection. Banks need to account for these problems while aiming to minimise losses in lending.

While it is accepted that all banks experience some loan losses, the degree of risk aversion varies significantly across institutions. All banks have their own **credit philosophy** established in a formal written **loan policy** that must be supported and communicated with an appropriate credit culture. The lending philosophy could reflect an emphasis on aggressive loan growth based on flexible underwriting standards. Alternatively, it could reflect the goals of a more conservative management aiming at achieving a consistent performance of a high-quality loan portfolio. Loan policies reflect the degree of risk bank management is ready to take and may change over time. A credit culture is successful when all employees in the bank are aligned with the management's lending priorities (Hempel and Simonson, 2008).

If internal data are available, credit risk can be monitored by looking at the changes in the ratio of medium-quality loans to total assets. The bank can choose to lower its credit risk by

BOX 11.1 CREDIT CULTURE AND EQUITY CULTURE IN BANKING

Credit culture refers to the fundamental principles that drive a bank's lending activity and the way management analyses risk. Credit culture relates specifically to the way management sets the values of a bank in relation to the credit risk management process and seeks to communicate with and influence the organisation. In this context, senior executives and particularly the CEO (chief executive officer) play a key role in determining a bank's credit culture and in setting the tone and direction of their organisations.

This special role of the CEO was also emphasised in an unpublished speech at Bangor Business School (September 2010) by Lord Mervyn Davies, former Government Minister and Chairperson of Standard Chartered Bank Plc. Lord Davies stressed the importance of the functions and responsibilities of a bank CEO in giving clear signals appropriate for the credit culture that they have decided to pursue. In line with the existing relevant literature, Lord Davies confirmed the role of the CEO as protector of a bank's value system, and in preserving prudent discipline, guidelines, and accountability.

In Mueller's (1994) words, credit culture can be defined as the true 'spirit behind the rules [...] with many moving parts embracing everything that has to do with credit extension. It is rooted in ideas, traditions, skills, attitude, philosophies and standards.' So credit culture not only relates to the tangible written policies and procedures, but also intangible. In addition, 'a credit culture is developed over time and communicated and passed on'. That is, a strong credit culture will permeate the organisation throughout and people will know intuitively what is acceptable and what is not.

The 2007–2009 financial turmoil demonstrated that an equity culture in banking took over from the credit culture in the run up to the collapse. Equity culture implies a bank's behaviour that encourages excessive risk taking and leverage and is characterised by short-term (and short-sighted) strategies. In a study published in the *OECD Financial Market Trends* paper series, Blundell-Wignall *et al*. (2009) attempt to identify the main factors that brought about a greater focus on equity culture at the expense of the credit culture. These include factors such as excess liquidity, poor regulation, competition and governance frameworks, structured products and derivative growth drivers, often motivated by tax considerations.

One way to 'diagnose' credit culture is to use the organisation behaviour approach, called the competing values framework (CVF). This conceptual model is based on four categories of activities that firms can use to create value, namely: (i) collaborate (investments in human capital); (ii) control (investments in internal processes); (iii) compete (activities to improve competitiveness); and (iv) create (organic innovation). More details on the CVF and how it can be employed to identify banks' credit culture can be found in Thakor (2016).

Sources: Blundell-Wignall *et al*. (2009); Davies (2010); Mueller (1994); Thakor (2016).

lowering this ratio. If the data on medium-quality loans are not available, traditional proxies for credit risk include, for instance:

- total loans/total assets;
- non-performing loans/total loans;
- loan losses/total loans;
- loan loss reserves/total assets.

These types of ratios can be calculated for different types of loans the bank holds on its balance sheet, for example, a bank may look at its mortgage loan book and see what proportion of such loans it holds relative to total assets, the amount on non-performing mortgage loans, loan losses on mortgage loans and so on.

However, Hempel and Simonson (2008) argue that these indicators can be subject to criticism because they lag in time behind the returns gained by taking higher risk. Therefore one should look at lead indicators such as:

- loan concentration in geographic areas or sectors;
- rapid loan growth;
- high lending rates;
- loan loss reserves/non-performing loans.

Another key credit-risk measure is the ratio of total loans to total deposits. The higher the ratio the greater the concerns of regulatory authorities, as loans are among the riskiest of bank assets. A greater level of non-performing loans to deposits could also generate greater risk for depositors.

Since in the presence of credit risk the mean return on the banks' asset portfolio is lower than if the loan portfolio was entirely risk-free, the bank will find it necessary to minimise the probability of bad outcomes in the portfolio by using a diversification strategy. Diversification will decrease **unsystematic (or firm-specific) credit risk**. This is derived from 'micro' factors and thus is the credit risk specific to the holding of loans or bonds of a particular firm. While diversification decreases firm-specific credit risk, banks remain exposed to **systematic credit risk**. This is the risk associated with the possibility that default of all firms may increase over a given period because of economic changes or other events that have an impact on large sections of the economy/market (e.g. in an economic recession firms are less likely to be able to repay their debts).

11.3 Interest rate risk

An interest rate is a *price* that relates to *present* claims on resources relative to *future* claims on resources. An interest rate is the price that a borrower pays in order to be able to consume resources now rather than at a point in the future. Correspondingly, it is the price that a *lender* receives to forgo current consumption. Like all prices in free markets, interest rates are established by the interaction of supply and demand; in this context, it is the supply of future claims on resources interacting with the demand for future claims on resources. An interest rate may, therefore, be defined as *a price established by the interaction of the supply of, and the demand for, future claims on resources*. That price will usually be expressed as a proportion of the sum borrowed or lent over a given period.

Interest rates have a crucial role in the financial system. For example, they influence financial flows within the economy, the distribution of wealth, capital investment and the profitability of financial institutions. For banks, the exposure to interest rate risk, that is the risk associated with unexpected changes in interest rates, has grown sharply in recent years as a result of the increased volatility in market interest rates especially at the international level.[1]

However, not all banks' assets and liabilities are subject to **interest rate risk** in the same way. Important distinctions should be made between **fixed-rate assets and liabilities** and **rate-sensitive assets and liabilities**:

- Fixed-rate assets and liabilities carry rates that are constant throughout a certain period (e.g. 1 year) and their cash flows do not change unless there is a default, early withdrawal or an unanticipated pre-payment.
- Rate-sensitive assets and liabilities can be re-priced within a certain period (e.g. 90 days); therefore the cash flows associated with rate-sensitive contracts vary with changes in interest rates.

Other banks' assets and liabilities can be categorised into non-earning assets (i.e. assets that generate no explicit income such as cash) and non-paying liabilities (i.e. liabilities that pay no interest, such as current accounts or, as they are known in the United States, 'checking accounts').

A rise in market interest rates has the effect of increasing banks' funding costs because the cost of variable rate deposits and other variable rate financing increases. If loans have been made at fixed interest rates this obviously reduces the net returns on such loans. Meanwhile, banks will be vulnerable to falling rates if they hold an excess of fixed-rate liabilities. In the case of bonds, as interest rates increase this reduces the market value of a bond investment. Typically, long-term, fixed-income securities subject their holders to the greatest amount of interest rate risk (see Appendix A1 for more details on the relation between bond prices and interest rates). In contrast, short-term securities, such as Treasury bills (T-bills), are much less influenced by interest rate movements.

Recently, regulators and bankers have been assessing the impact on loan revenues and bank margins of persistently low interest rates and the prospect of it continuing at least until 2023, as shown in the article reported in Box 11.2.

Traditional interest rate risk analysis compares the sensitivity of interest income to changes in asset yields with the sensitivity of interest expenses to changes in interest costs of liabilities. In particular, it is common to refer to the ratio of *rate-sensitive assets* to *rate-sensitive liabilities*: when rate-sensitive assets exceed rate-sensitive liabilities (in a particular maturity range), a bank is vulnerable to losses from falling interest rates. Conversely, when rate-sensitive liabilities exceed rate-sensitive assets, losses are likely to be incurred if market interest rates rise. Typically, if a bank has a ratio above 1.0, the bank's returns will be lower if interest rates decline and higher if they increase. However, given the difficulty in forecasting interest rates, some banks conclude that they can minimise interest rate risk with an interest sensitivity ratio

[1] Note that interest rate risk is faced by both borrowers (who do not want to pay higher rates on loans) and lenders who may suffer losses if rates fall.

BOX 11.2 US BANKS TRIM EXPECTATIONS IN ERA OF LOW INTEREST RATES

US banks are preparing investors for a prolonged period in which low interest rates are a drag on their profits.

Speaking at an industry conference last week, most of the nation's biggest banks — including JPMorgan Chase, Wells Fargo, Citigroup, and Bank of America — trimmed their forecasts for net lending revenues for 2020, attributing the cuts to sustained low rates as well as waves of homeowners refinancing their mortgages at lower rates.

JPMorgan, for example, cut its net interest income forecast by $1bn, to $55bn — as compared to $57bn in 2019.

The US Federal Reserve projected last week that rates would remain at their current rock-bottom level until at least 2023, and that it would not tighten policy without first seeing a sustained period of inflation. This is bad news for banks' profit margins, as low interest rates depress loan yields and the cost of deposit funding cannot fall much further.

Worries about the impact of low rates drowned out more encouraging news delivered in the banks' updates. Most said that reserves for credit losses, after big increases in the first two quarters of the year, were unlikely to rise in the third, given solid credit performance. Almost every bank reported that the number of borrowers taking advantage of payment holidays continued to fall.

"We set our second-quarter reserves with a set of scenarios [and it] turned out that the actual data has been better ... charge-offs keep coming down, frankly, because of credit quality," said Bank of America chief executive Brian Moynihan. Wells Fargo chief financial officer John Shrewsberry noted that in commercial lending "we've actually seen some better realised outcomes than we imagined". Many other executives echoed that theme.

Banks' provisions for bad loans have risen by $111bn this year, to $223bn, close to the peak levels of the financial crisis, according to the Fed.

"The group is going down two different tracks — credit and rates," summed up Scott Siefers, bank analyst at Sandler O'Neill. While the rate environment is a drag, "the credit updates have been about as good as can be hoped. The big reserves are done, at least for now. The [loan payment] deferral updates were constructive too." Mr Siefers said the emerging consensus on Wall Street was that this would be a "confined" credit cycle, with certain industries hit hard, but most performing reasonably well.

Another piece of good news: banks' capital markets operations have continued to benefit from higher activity. Both JPMorgan, Citi and BofA for example, project double-digit year-over-year increases in trading revenues for the third quarter.

Yet markets continue to focus resolutely on the bad news. Banks' stocks have underperformed the wider market by 30 per cent since the Covid-19 crisis began and trade at whopping valuation discounts. The S&P 500 now has a price/earnings multiple of 24; many banks trade at 10 to 12 times earnings.

The low expectations continue into next year: analysts' estimates for the large banks' earnings per share in 2021 have not recovered at all from the lows they hit a few months ago, and are a third or more below where they were in February.

The pervasive pessimism has left bank investors scratching their heads. Eric Hagemann, of Pzena Investment Management — which manages $35bn and has substantial positions in banks including JPMorgan, Citi and Wells Fargo — said that banks' widening discount made little sense, given that low interest rates should support all stock valuations, by lowering the discount rate for future profits.

"Banks are underperforming because of low rates, but low rates should increase the multiple of all equities. Earnings-per-share estimates coming down makes sense — but multiple contraction, too? It seems like double punishment." He also noted the potential for some banks' heavy provisions for bad loans being released as the crisis abates. "If there are reserve releases, the earnings numbers these guys are going to put are going to be huge," he said.

FT *Source*: Armstrong, R. (2020) 'US banks trim expectations in era of low interest rates', *Financial Times*, 20 September. © The Financial Times Limited 2020. All Rights Reserved.

close to 1.0. Such a ratio may be difficult for some banks to achieve and often can be reached only at the cost of lower returns on assets.

These traditional measures of interest rate risk have a number of limitations. Bank managers nowadays use sophisticated measures of interest rate management such as the gap buckets analysis, maturity models and duration analysis. Details of these models are provided in Chapter 12. It is important to note that interest rate risk measurement and management will always imply an ALM approach (see Chapter 10).

Section 11.3.1 follows Saunders and Cornett's (2017) examples of the two types of interest rate risk that banks may have to face – **refinancing risk** and **reinvestment risk** – and also focuses on the impact on a bank's profitability.

11.3.1 Refinancing risk and reinvestment risk

Interest rate risk is the risk arising from the mismatching of the maturity and the volume of banks' assets and liabilities as part of their asset transformation function. As we saw in Chapter 1, one of the main functions of financial intermediaries is to act as asset transformers, e.g. they transform short-term deposits into long-term loans.

Typically the maturity of banks' assets (e.g. loans) is longer than the maturity of banks' liabilities (e.g. deposits). This means that banks can be viewed as 'short-funded' and have to run the risk of refinancing their assets at a rate that may be less advantageous than previous rates. Saunders and Cornett (2017) define refinancing risk as: 'the risk that (the) cost of rolling over or re-borrowing funds will rise above the returns being earned on asset investments'.

For example, assume the cost of funds (liabilities) for a bank is 7 per cent per annum and the interest return on an asset is 9 per cent. In year 1 by borrowing short (1 year) and lending long (2 years) the profit spread is equal to the difference between the lending and borrowing rates, i.e. 2 per cent, as in situation (a) in Figure 11.1. In year 2 profits are uncertain because if the interest rate does not change then the bank can refinance its liabilities at 7 per cent per annum and as in year 1 have a profit of 2 per cent. However, if interest rates increase as in situation (b), then the bank can borrow at 8 per cent and profits will be only 1 per cent.

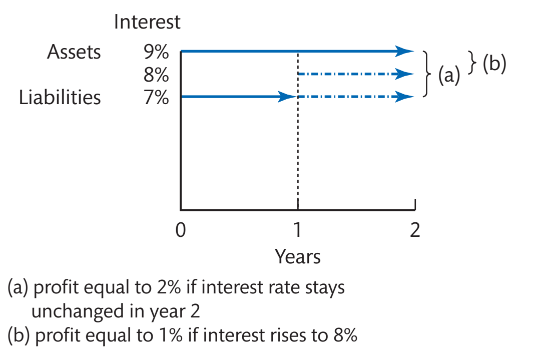

Figure 11.1 Refinancing risk

Source: Adapted from Saunders and Cornett (2017).

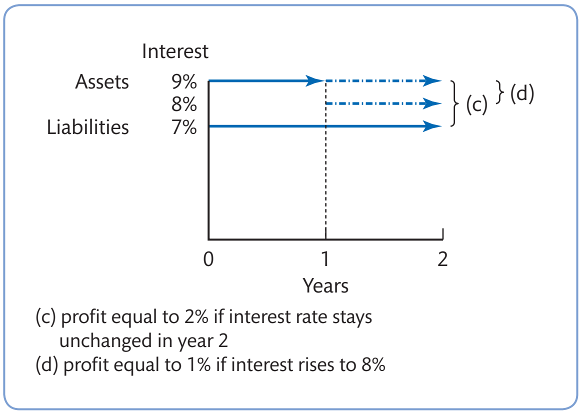

Figure 11.2 Reinvestment risk

Source: Adapted from Saunders and Cornett (2017).

Another type of mismatch could occur when the maturity of banks' liabilities is longer than the maturity of the assets. This means that banks can be viewed as 'long-funded', and have to run the risk of reinvesting their funds in the second period at a rate that may be less advantageous than the previous rate, and that they were unable to forecast in advance. Such a situation can often occur in banking within specific maturity buckets (i.e. intervals of consecutive maturities, such as a 3–6 months' maturity bucket). Accordingly, Saunders and Cornett (2017) define reinvestment risk as: 'the risk that the returns on funds to be reinvested will fall below the cost of funds'.

For example, assume the interest return on an asset for a bank is 9 per cent per annum and the cost of funds (liabilities) is 7 per cent. In year 1 by borrowing long (2 years) and lending short (1 year) the profit spread is equal to the difference between lending rate and borrowing rate, that is 2 per cent, as in situation (c) in Figure 11.2. In year 2 profits are uncertain because if the interest rate does not change then the bank can reinvest its assets at 9 per cent per annum and as in year 1 have a profit of 2 per cent. However, if interest rates go down as in situation (d), then the bank can lend at 8 per cent and profits will be only 1 per cent.

11.4 Liquidity (or funding) risk

A liquid asset may be defined as an asset that can be turned into cash quickly and without capital loss or interest penalty. Most bank deposits are therefore very liquid, but investment in, for example, property is highly illiquid. The liquidity that is required by a lender will depend on a number of factors, including in particular the range of liquidity inherent in other securities held. Moreover, a bank needs liquidity to cover a possible surge in operating expenses and to satisfy loan demand. All other things being equal, lenders will wish to have a high level of liquidity in their loans.

Liquidity risk is generated in the balance sheet by a mismatch between the size and maturity of assets and liabilities. It is the risk that the bank is holding insufficient liquid assets on its balance sheet and thus is unable to meet requirements without impairment to its financial or reputational capital. Banks have to manage their liquidity to ensure that both predictable and unpredictable liquidity demands are met and taking into account that the immediate sale of assets at low or 'fire sale' prices could threaten the bank's returns.[2]

However, liquidity risk can also occur when the duration of the balance sheet is matched: consider the situation if some of the loans funded by deposits were to default. If loan defaults were so large as to leave insufficient liquidity for the bank to satisfy depositors' demands to withdraw these deposits, then the bank would need to find new deposits to fund the withdrawals. As illustrated in the example in Box 11.3, information asymmetries between the bank and the depositors can often originate liquidity risk.

If a bank cannot meet depositor demands there will be a bank run, as depositors lose confidence and rush to withdraw funds. This may then make it difficult for the bank to obtain funds in the interbank market and before long a **liquidity crisis** will turn into a solvency crisis and possible failure (on the distinction between liquidity and solvency see Chapter 5, Box 5.6). Hence it is common to distinguish two types of liquidity risk:

- Day-to-day liquidity risk relates to daily withdrawals. This is usually predictable (or 'normal') because only a small percentage of a bank's deposits will be withdrawn on a given day. Very few institutions ever actually run out of cash because it is relatively easy for the bank to cover any shortage of cash by borrowing funds from other banks in the interbank markets.
- A liquidity crisis occurs when depositors demand larger withdrawals than normal. In this situation banks are forced to borrow funds at an elevated interest rate, higher than the

BOX 11.3 LIQUIDITY RISK AND INFORMATION ASYMMETRIES

Consider a bank that has made loans of €1 million with three-year maturity financed with uninsured demand deposits. The bank has an informational advantage compared with outsiders on the credit risk and overall quality of its loan portfolio. Suppose that one year later €500,000 of deposits are withdrawn and the bank's stock of cash assets is only €100,000. The bank will need €400,000 to fund the deposit withdrawal. Two cases can occur: (1) potential new depositors have a good perception about the bank's loan portfolio and the bank will easily acquire the €400,000 in new deposits; (2) outsiders have received unfavourable information about the bank's loans despite the bank's belief that the quality is good. Here the bank will either not be able to acquire new deposits (in extreme cases it will lose all existing deposits!) or have to pay too high a price for the risk associated with the loan portfolio. Informational asymmetries about asset quality create liquidity risk. If outsiders knew as much as the bank does about the quality of the loan portfolio, then the bank would be able to acquire deposits at the appropriate price given the risk of the loan portfolio.

Information asymmetries can also affect interbank lending and indeed evidence has shown that the actions taken during the 2007–2009 financial turmoil did little to restart the frozen interbank markets. These measures included central bank monetary policy decisions to lower rates, quantitative easing and bank recapitalisations. Each day banks exchange liquidity in the interbank markets and this process is crucial to guarantee a smooth functioning of monetary policy and for the healthy management of bank liquidity.

Source: Adapted from Greenbaum *et al*. (2019).

[2] Greenbaum *et al*. (2019) point out that the most extreme manifestation of liquidity risk is that the seller of the asset is *unable* to sell the asset at any price. This is known as *credit rationing* and occurs when a bank refuses credit to a borrower irrespective of the price they are willing to pay.

market rate that other banks are paying for similar borrowings. This is usually unpredictable (or 'abnormal') and can be due to either a lack of confidence in the specific bank, or some unexpected need for cash. Liquidity crises can ultimately hinder the ability of a bank to repay its obligations and in the absence of central bank intervention or deposit insurance it could result in 'a run' and even the insolvency of the bank.

Typically banks can reduce their exposure to liquidity risk by increasing the proportion of funds committed to cash and readily marketable assets, such as T-bills and other government securities, or use longer-term liabilities to fund the bank's operations. The difficulty for banks, however, is that liquid assets tend to yield low returns so if a bank holds sub-optimal levels of such assets its profits will decline. This is the trade-off between liquidity and profitability: the opportunity cost of stored liquidity is high and holding low-yielding assets (and/or zero-yielding assets such as cash) on the balance sheet reduces bank profitability (this issue will be considered further in Chapter 12).

One measure banks can use to monitor liquidity risk relates to short-term securities, a proxy for a bank's liquidity sources, to total deposits – this provides an approximate measure of a bank's liquidity needs. Another traditional ratio of liquidity risk is the loan/deposits ratio. This ratio tends to focus on the liquidity of assets on the balance sheet. A high ratio of short-term securities/deposits and a low loan/deposits ratio indicate that the bank is less risky but also less profitable.

There are, however, other indicators that are more suited to proxy for a bank's liquidity needs, based on actual or potential cash flows. For example, a good indication of the bank's need for liquidity may be given by the amount a bank has in purchased or volatile funds and the amount the bank has used of its potential borrowing reserve.

Liquidity shortage can potentially cause a credit crunch (or credit squeeze) but usually actions are taken by central banks to avoid or limit the reduced availability of credit in the system. Recently, central banks around the world went long lengths to support the markets and keep them functioning during the Covid-19 outbreak. For example, in March 2020, when the US treasury bond market was dysfunctional and lost its depth and liquidity, the Fed launched a series of actions to stabilise the treasury markets and provided extraordinary liquidity support (see Box 11.4).

BOX 11.4 THE FEDERAL RESERVE HAS GONE WELL PAST THE POINT OF 'QE INFINITY'

The vast and currently dysfunctional markets for US Treasuries, mortgages and corporate credit now have the ultimate buyer of last resort — the Federal Reserve.

Ever since the big stock market crash of 1987, investors have grown to depend on the US central bank coming to the aid of financial markets when they hit the skids. Now the central bank is well and truly "all in", announcing a slew of new initiatives on Monday designed to buy time for an enfeebled financial system. Until now, the Fed implied, the system has been in no shape to withstand an escalating pandemic that has the US economy facing the biggest hit to growth since the 1930s.

Having failed to ease nerves over the past week with an expansion of its quantitative easing programme, the Fed has upped the ante to Buzz Lightyear territory. "QE infinity", in the form of unlimited buying of Treasury debt and

BOX 11.4 The Federal Reserve has gone well past the point of 'QE infinity' (*continued*)

mortgage-backed securities, is just one aspect of the new approach.

Another notable measure is the Fed's entry into the universe of investment-grade credit, with the central bank launching a facility that will enable the purchase of corporate debt that has already been sold with a maturity of less than five years, and also of exchange traded funds that track the sector. Yet another facility will support new sales of bonds and loans from companies with investment-grade credit ratings. One way of looking at this is to note that the Fed has joined the Bank of Japan, the European Central Bank and the Bank of England in supporting credit markets. Viewed another way, it complicates any future exit strategy and extends a long post-Lehman crisis legacy of distorted risk premia in markets.

But perhaps now is not the time to quibble over legacies, while the pandemic rages. The purchase of corporate debt by the Fed is designed to prevent a deeper and prolonged bust facing an economy that has been placed in suspended animation. A highly indebted US corporate sector, fuelled by Fed policy objectives since the last crisis, is being stripped of earnings growth and cash flows for an indeterminate period. This is led by companies operating in the travel, leisure, restaurant and retail sectors, and sets the stage for a deepening of the current macro-economic shock.

Credit risk premia have moved sharply wider in recent weeks, catching out many investors, while also preventing lower quality borrowers from refinancing their maturing debts in the form of commercial paper and corporate bonds. Losses for many fund managers are mounting, while exchange traded funds have become highly volatile and dysfunctional. In the US Treasuries market, meanwhile, when liquidity dries up, conditions for corporate credit become even more stressed, intensifying a fire-sale environment as investors rush for exits and seek to cash out at any price.

An aggressive expansion of QE by the Fed has loomed large since the eruption of market turmoil just over a month ago. Some have long argued that unlimited quantitative easing was a likely final step, after the policy appeared to become a permanent feature of the Fed's toolkit in the wake of the financial crisis.

The policy of suppressing long-dated Treasury yields — and by extension, broader market volatility — was billed as a temporary measure when introduced in 2009. But it has pretty much become business as usual since then for the Fed and other leading central banks. Efforts by the Fed to raise interest rates by modest amounts, and to trim the balance sheet, were terminated after severe market turmoil erupted in late 2018, casting a lengthy shadow over business and consumer confidence. A financial system replete with debt was in no fit state to handle even a slight rise in Treasury yields back then.

Now, an even more indebted corporate sector faces intense pressure — until a peak in the spread of infections allows the economy to start rebounding, unblocking cash flows and boosting earnings to some degree. The ultra-low borrowing costs of recent years provided companies with licence to tap credit markets to fund expansive share buyback programmes that benefited executives. The failure to save for a rainy day means many companies are now holding out for help from the taxpayer.

The Fed's new set of measures also means that the central bank's balance sheet will increase markedly in size as it becomes ensconced as the buyer of last resort across fixed-income markets. Investors should not expect any contraction any time soon, given the scale of the debt binge since 2009.

And they should not rule out further support measures, should markets remain treacherous and hard to trade. While the Fed's intervention boosted some areas of the US credit market on Monday, stocks were muted, reflecting the other game in town — negotiations over a $2tn fiscal response to the economic effects of the health crisis.

FT *Source*: Mackenzie, M. (2020) 'The Federal Reserve has gone well past the point of 'QE infinity', *Financial Times*, 23 March.

11.5 Foreign exchange risk

As banking markets become more global, the importance of international activities in the form of foreign direct investment and foreign portfolio investments has increased sharply. However, the actual return banks earn on foreign investment may be altered by changes in exchange rates. Changes in the value of a country's currency relative to other currencies affect the foreign exchange rates. Like other prices, exchange rates (which essentially reflect the price of currencies) tend to vary under supply and demand pressure.

Foreign exchange relates to money denominated in the currency of another nation or group of nations. Any firm or individual that exchanges money denominated in the 'home' nation's currency for money denominated in another nation's currency can be said to be acquiring foreign exchange. This is the case whether the transaction is very small, involving, say, a few pounds, or whether it is a company changing a billion dollars for the purchase of a foreign company. In addition, the transaction is also viewed as acquiring foreign exchange if the type of money being acquired is in the form of foreign currency notes, foreign currency bank deposits, or any other claims that are denominated in foreign currency. Put simply, a foreign exchange transaction represents a movement of funds, or other short-term financial claims, from one country and currency to another.

Foreign exchange can take many forms. It can be in the form of cash, funds available on credit cards (credit card payments when on holiday overseas are usually made in the foreign currency, but the card is debited in home currency), bank deposits or various other short-term claims. In general, a financial claim can be regarded as foreign exchange if it is negotiable and denominated in a currency other than that in which it resides, for instance, a US dollar bank deposit in Paris. Box 11.5 provides a more formal definition and explanation of exchange rates and foreign exchange markets.

BOX 11.5 EXCHANGE RATES AND FOREIGN EXCHANGE MARKETS

What is an exchange rate?

The exchange rate is a *price* – the number of units of one nation's currency that need to be surrendered in order to acquire one unit of another nation's currency. There is a wide range of 'exchange rates', for example the US dollar, Japanese Yen, British pound and the Euro. In the spot market, there is an exchange rate for every other national currency traded in that market, as well as for a variety of composite currencies such as the International Monetary Fund's Special Drawing Rights 'SDRs'. There are also a variety of 'trade-weighted' or 'effective' rates designed to show a currency's movements against an average of various other currencies.

In addition to the spot rates, there is also a wide range of additional exchange rates for transactions that take place on other delivery dates, in the forward markets. While one can talk about, say, the Euro exchange rate in the market, it is important to note that there is no single, or unique Euro exchange rate in the market, as the rate depends on when the transaction is to take place, the spot rate being the current rate and forward rates being influenced by various factors most noticeably the Euro interest rate compared to the official interest rate in other currencies.

The market price is determined by supply and demand factors, namely, the interaction of buyers and sellers, and the market rate between two currencies is determined by the interaction of the official (mainly central banks) and private participants (banks, companies, investment firms, and so on) in the foreign exchange market. For a currency with an

BOX 11.5 Exchange rates and foreign exchange markets (*continued*)

exchange rate that is set by the monetary authorities, the central bank or another official body is the key participant, standing ready to buy or sell the currency as necessary to maintain the authorised rate.

What are the features of the foreign exchange markets?

While most people are familiar with the aforementioned features of foreign exchange they are less familiar with what is known as the foreign exchange market. The foreign exchange market represents an international network of major foreign exchange dealers (central banks, commercial banks, brokers and other operators) that undertake high-volume (wholesale) trading around the world. These transactions nearly always take the form of an exchange of bank deposits of different national currency denominations. If one bank agrees to sell euro for US dollars to another bank, there will be an exchange between the two parties of a euro bank deposit for a US dollar bank deposit.

There is no physical location for the foreign exchange market, and transactions are carried out via computer screens based in banks, large companies and various other organisations throughout the world. The majority of foreign exchange turnover takes place in the main financial centres of London, New York and Tokyo. The main market participants include banks, non-financial firms, individuals, official bodies and other private institutions that are buying and selling foreign exchange at any particular time. Some of the buyers and sellers of foreign exchange may be involved in physical goods transactions and need foreign exchange to make purchases (such as a UK importer of car parts from Japan that needs to make a purchase in yen). However, the proportion of foreign exchange transactions related to goods trade is generally believed to be very small – less than 5% of all foreign exchange transactions. Other participants in the market may wish to undertake foreign exchange transactions to undertake direct investment in plant and equipment, or in portfolio investment (dealing across borders in stocks and bonds and other financial assets), while others may operate in the money market (trading short-term debt instruments internationally).

Overall, the motives of participants in the foreign exchange market are wide and varied. Some participants are international investors or speculators while others may be financing foreign investments or trade. Many participants also use foreign exchange transactions for risk management purposes in order to hedge against the risks associated with adverse foreign exchange movements. Transactions may be very short-term or long-term, and can be conducted by official bodies (such as central banks) or by private institutions, the motives for conducting business can vary and the scale of activity can alter substantially over time. All these combined features make up the supply and demand characteristics of the global foreign exchange market.

Given the diverse make-up of the supply and demand features of the market, predicting the future course of exchange rates is a particularly complex and uncertain business. In addition, since exchange rates influence such an extensive array of participants and business decisions, it is a critically important price in an economy, influencing consumer prices, investment decisions, interest rates, economic growth, the location of industry, and much more. It should be clear that for a firm wishing to undertake international activity knowledge of the foreign exchange market, predictions about future exchange rates and the risks associated with foreign exchange are essential elements that senior managers must understand if they are to be successful in overseas expansion.

The foreign exchange market consists of a wholesale (or interbank) market and a retail market. Transactions in the wholesale market are dominated by bank-to-bank transactions that involve very large transactions, typically greater than $1 million. In contrast, the retail market, where clients obtain foreign exchange via their banks involves much smaller sums.

According to the 2019 Triennial Survey by the Bank for International Settlements (BIS), trading in foreign exchange market reached $6.6 trillion per day in April 2019 compared to $5.1 trillion in 2016. FX swaps were the most actively traded instruments in April 2019, at $3.2 trillion per day accounting for 49 per cent of total FX market turnover. The dominant currency status was the US dollar, being on one side of 88 per cent of all trades followed by the euro (32 per cent) and the Japanese yen (17 per cent).

Source: Bank for International Settlements (2019).

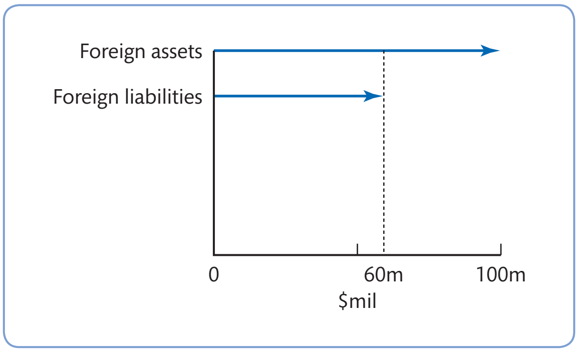

Figure 11.3 The foreign asset and liability position of a Spanish bank: a net long asset position in US$

Source: Adapted from Saunders and Cornett (2017).

Foreign exchange risk is the risk that exchange rate fluctuations affect the value of a bank's assets, liabilities and off-balance-sheet activities denominated in foreign currency. We have seen in Chapter 9 that UK banks have significant amounts of foreign currency assets and liabilities in their balance sheet – this is mainly from the assets and liabilities of foreign banks based in London that do predominantly wholesale foreign currency-related business (see Tables 9.2 and 9.3). A bank may be willing to take advantage of differing interest rates or margins in another country, or simply to invest abroad in a currency different from the domestic one. A bank that lends in a currency that depreciates more quickly than its home currency will be subject to foreign exchange risk.

Suppose a Spanish bank has a net 'long' assets position in dollars (e.g. a loan in dollars) of $100 million as illustrated in Figure 11.3. (If the bank is *net long* in foreign assets it means it holds more foreign assets than liabilities. Conversely, the bank would be holding a *net short* position in foreign assets if it had more foreign liabilities than assets.) On the liability side, the Spanish bank has $60 million in Spanish CDs. These CDs are denominated in euros. The Spanish bank will suffer losses if the exchange rate for dollars falls or depreciates against the euro over this period because the value of the US$ loan assets would decrease in value by more than the euro CDs.

To measure foreign exchange risk banks calculate measures of net exposure by each currency. It will be equal to the difference between the assets and liabilities denominated in the same currency. In the case above the bank is exposed to the risk that its net foreign assets may have to be liquidated at an exchange rate lower than the one that existed when the bank entered into the foreign asset/liability position.

11.6 Market (or trading) risk

Market risk is the risk of losses in on- and off-balance-sheet positions arising from movements in market prices. It pertains in particular to short-term trading in assets, liabilities and derivative products, and relates to changes in interest rates, exchange rates and other asset prices.

Modern conditions have led to an increase in market risk due to a decline in traditional sources of income and a greater reliance by banks on income from trading securities. This process has increased the variability in banks' earnings due to the relatively frequent changes in market conditions. International regulators have acknowledged the importance of market risk since 1996 when the Basel Capital Accord was changed to incorporate capital requirements for market risk in its capital adequacy rules (see Chapter 7).

Heffernan (2005) distinguishes between:

- general or systematic market risk, caused by a movement in the prices of all market instruments due to macrofactors (e.g. a change in economic policy); and
- unsystematic or specific market risk, which arises in situations where the price of one instrument moves out of line with other similar instruments because of events related to the issuer of the instrument (e.g. an environmental law suit against a firm will reduce its share prices but is unlikely to cause a decline in the market index).

Market risk is the risk common to an entire class of assets or liabilities. It is the risk that the value of investments may decline over a given period simply because of economic changes or other events that affect large portions of the market. Typically, market risk relates to changes in interest rates, exchange rates and securities' prices driven by the market overall. In the case of bank lending, credit risk is the most important, but for banks lending to companies that are investing in securities (such as bank loans to hedge funds) then bank assessment of credit risk will be influenced by the hedge funds' exposure to market risk. (The banks' own investments and securities trading activity will also be subject to market risk.)

Bonds and shareholders' equity are especially sensitive to the movements in market interest rates and currency prices and this affects the investors' perception of a bank's risk exposure and earnings potential. Important indicators of market (or price) risk in banking are (Rose and Hudgins, 2010):

- book value of assets/estimated market value of those same assets;
- book value of equity capital/market value of equity capital;
- market value of bonds and other fixed-income assets/their value as recorded on a financial institution's book;
- market value of common and preferred stock per share, reflecting investor perceptions of a financial institution's risk exposure and earnings potential.

Large banks typically perform value-at-risk (VaR) analysis to assess the risk of loss on their portfolios of trading assets while small banks measure market risk by conducting sensitivity analysis. VaR is a technique that uses statistical analysis of historical market trends and volatilities to estimate the likely or expected maximum loss on a bank's portfolio or line of business over a set period, with a given probability. The aim is to get one figure that summarises the maximum loss faced by the bank within a statistical confidence interval. For instance, it can be estimated that 'there is a 0.5 per cent chance that a portfolio will lose £1 million in value over the next quarter'. Another measure described in the new framework (see Box 11.6) is expected shortfall (ES). It is calculated as the average of all potential losses exceeding the VaR at a given confidence level, which makes up for VaR's shortcomings in capturing the risk of extreme losses. In contrast, sensitivity analysis refers to the more traditional methods of assessing the price sensitivity of assets and liabilities to changes in interest or other rates and the corresponding impact on stockholders' equity.

BOX 11.6 A REVISED FRAMEWORK FOR MARKET RISK

Market risk emphasises the risks for banks that trade their assets, liabilities and derivative contracts actively and frequently in the markets. For regulatory purposes these highly liquid positions in different financial instruments (such as for example, bonds, equities and derivatives) typically belong to the **trading book** of the bank. In contrast, the **banking book** includes deposits, cash, loans and other illiquid assets as shown in Figure 11.4.

Although the nature of the activities reported in the trading and banking book is not fundamentally different, they are subject to separate regulatory treatment. In general, market-based activities are less *onerous* in terms of capital requirements and this has generated distortions in the system by incentivizing banks to move items from the banking book to the trading book thereby adding risks. In other words, the lack of precise boundaries between banks' *banking books* and *trading books* has resulted in regulatory arbitrage and inadequate capital levels with regards to the trading books.

In 2012 the Basel Committee started a review on the design of the trading book regime with the aim to address the shortcomings of both standardized and internal model approaches. The revised market risk framework was published in January 2019. In addition to the boundary of the banking book and the trading book, the new framework includes modifications concerning the internal models approach and the standardised approach for the management of market risk.

Source: Basel Committee on Banking Supervision (2019a).

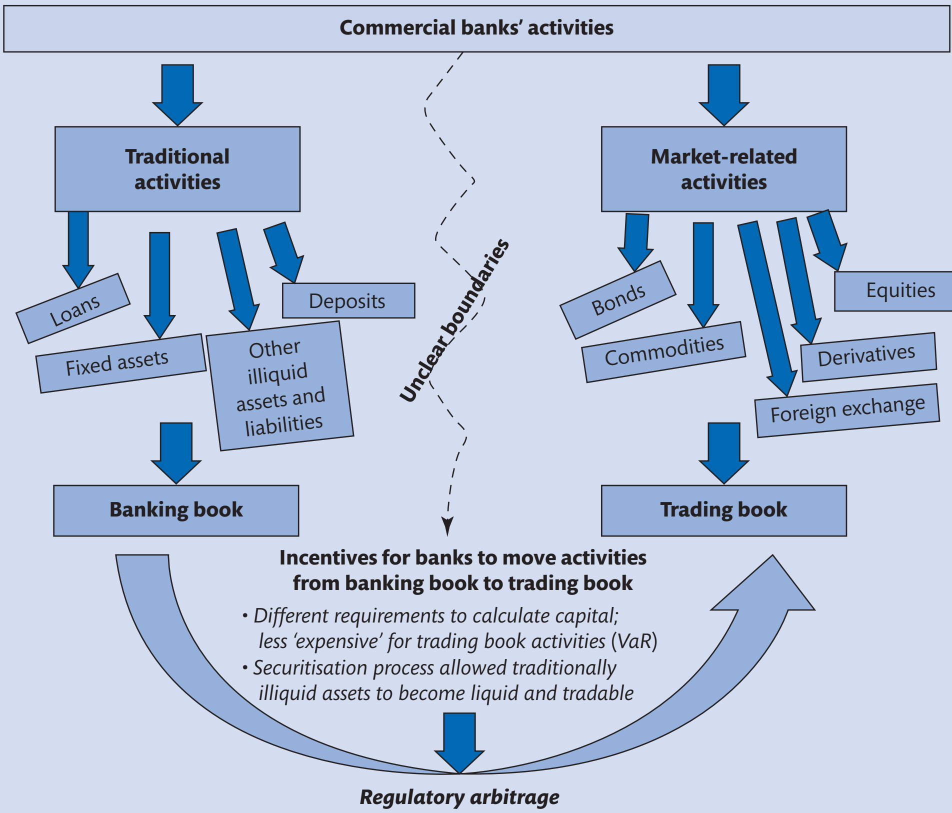

Figure 11.4 Banking vs. trading books

Source: Basel Committee on Banking Supervision (2013c).

11.7 Country and sovereign risk

Country risk is the risk that economic, social and political conditions of a foreign country will adversely affect a bank's commercial and financial interests. It relates to the adverse effect that deteriorating macroeconomic conditions and political and social instability may have on the returns generated from overseas investments. International lending always carries 'unusual' risks; however, it is generally accepted that a loan to a foreign government is safer than a loan to a private sector borrower (Hempel and Simonson, 2008).

Nonetheless, it is possible that even governments will default on debt owed to a bank or government agency. This is the **sovereign risk** and refers to the possibility that governments, as sovereign powers, may enforce their authority to declare debt to external lenders void or modify the movements of profits, interest and capital. Such situations typically arise when foreign governments experience some sort of economic or political pressures and decide to divert resources to the correction of their domestic problems. Clearly the fact that governments generally benefit from a particular 'immunity' from legal process makes international lending extremely risky for banking institutions that in case of default find it very difficult, if not impossible, to recover some of the debt by taking over some of the country's assets.

The Basel Committee (BCBS, 2018) identifies five sources of sovereign risk and five channels through which it can affect the banking system, either in isolation or in combination. The sources are (i) missed payments, (ii) debt restructuring or outright defaults, (iii) currency redenomination, (iii) currency devaluations, (iv) losses from unanticipated, higher inflation and (v) fluctuations in the value of sovereign exposures. The channels are:

- direct exposure ($\uparrow$ sovereign risk = $\downarrow$ the value of banks' sovereign exposures);
- collateral ($\uparrow$ sovereign risk = $\downarrow$ the value of sovereign collateral used by banks);
- sovereign credit rating downgrades ($\uparrow$ sovereign risk = $\downarrow$ downgrades to the ratings of other entities in the economy);
- government support ($\uparrow$ sovereign risk = $\downarrow$ funding benefits that banks derive from government guarantees);
- macroeconomic ($\uparrow$ sovereign risk = $\uparrow$ credit tightening due to greater borrowers riskiness, banks' fragility and funding costs).

Sovereign risk can result in the re-scheduling and re-negotiation of the debt, with considerable losses for the lending banks. These types of new agreements are usually obtained with the intervention of international organisations (e.g. the International Monetary Fund and the World Bank). The extreme case for sovereign risk is *debt repudiation*, when the government simply repudiate their debts and no longer recognise their obligations to external creditors.

The channels listed above can increase banks' susceptibility to distress or even failure, and can also raise the risk of contagion to the rest of the banking sector. However, there can also be reverse causality running from the banking sector (e.g. a banking crisis) to sovereign fragility. This is because banks and sovereigns are highly interconnected (the so-called sovereign–bank nexus) and problems originating in one of the two sectors can cause an adverse

BOX 11.7 EUROPEAN BANKS LOAD UP ON GOVERNMENT BONDS, RAISING CONCERNS OVER 'DOOM LOOP'

European banks have loaded up on more than €200bn of their own governments' bonds since the start of the Covid-19 pandemic, in a move that could reawaken fears about the sector's growing stockpiles of risky sovereign debt.

According to research by S&P Global Ratings, banks had increased their holdings of home-country government bonds to nearly €1.6tn by the end of June, up 15 per cent from the end of February. The rating agency said the pace of purchases was seven times faster than in the same period in 2019.

S&P said banks in the eurozone's former crisis-hit "periphery" — such as Italy, Greece and Spain — have the highest exposure to their governments' debt, along with those outside the euro area in central and eastern Europe.

The figures are likely to revive concerns about the effect that wild swings in the price of government bonds could have on banks' balance sheets. During the eurozone crisis, sell-offs in sovereign debt repeatedly dragged down profits and share prices in the banking sector, in turn raising the prospect of a further hit to the economy — a dynamic labelled the "doom loop".

"The surge in home-government bond investment in Europe is a temporary response to excess market liquidity, in our view. We think this time is different than in the run-up to the European sovereign debt crisis starting about 2011," said Cihan Duran, a credit analyst at S&P Global Ratings.

"If this turns out not to be true and the trend persists in Europe, overlooking sovereign risks could unleash a new 'doom loop' in the distant future, particularly where banks have built up extremely large exposures to sovereign debt," he said.

The spread of the pandemic in March triggered a sell-off in global bond markets, which was particularly acute in the countries on the periphery of the eurozone. The European Central Bank helped to avert a new debt crisis with a €750bn emergency asset purchase programme, which was later expanded to €1.35tn in June. The central bank followed up with a fresh round of super-cheap loans to banks, resulting in a rush to borrow €1.3tn at negative rates.

While the injection of cash has increased the financial sectors' ability to lend to households and businesses, much of it has ended up parked in sovereign debt, according to S&P. Banks in the EU do not have to hold any capital against their own governments' bonds, which are considered a "risk-free" investment by regulators.

"We believe the central bank's policies are unintentionally acting as an incentive for banks to add to their sovereign debt holdings. The ECB may take additional accommodative measures that might accelerate this trend," the report said.

However S&P added that the huge run-up in government bond holdings is likely to unwind once the recovery from the Covid crisis kicks in.

FT *Source*: Stubbington, T. (2020) 'European banks load up on government bonds, raising concerns over "doom loop"', *Financial Times*, 21 September.

'feedback loop' that tends to amplify the effects in both sectors. This feedback loop occurred during the European sovereign debt crisis and is again causing concern in European banks during the Covid-19 pandemic, as shown in Box 11.7.

To help investors measure country and sovereign risks, rating company agencies provide tables of sovereign (credit risk) ratings. These ratings typically act as a rating 'ceiling' for

Table 11.1 Selected sovereign foreign currency ratings

Country	Rating	Country	Rating	Country	Rating
Argentina	CCC+	Hungary	BBB	Poland	A−
Australia	AAA(−)	Iceland	A	Portugal	BBB
Austria	AA+	India	BBB−	Qatar	AA−
Belgium	AA	Indonesia	BBB(−)	Russia	BBB−
Brazil	BB−	Ireland	AA−	Singapore	AAA
Bulgaria	BBB(+)	Israel	AA−	Slovakia	A+
Canada	AAA	Italy	BBB−	South Africa	BB−
Chile	A+(−)	Japan	A+(+)	South Korea	AA
China	A+	Jordan	B+	Spain	A
Colombia	BBB−(−)	Kenya	B+	Sweden	AAA
Croatia	BBB−	Malaysia	A−	Switzerland	AAA
Cyprus	BBB−	Mexico	BBB(−)	Taiwan, China	AA−
Czech Rep.	AA−	Netherlands	AAA	Thailand	BBB+
Denmark	AAA	New Zealand	AA(+)	Turkey	B+
Finland	AA+	Norway	AAA	United Kingdom	AA
France	AA	Pakistan	B−	United States	AA+
Germany	AAA	Peru	BBB+	Venezuela	SD
Greece	BB−	Philippines	BBB+	Zambia	SD

Note: The best-quality credits are rated AAA ('prime'), followed by AA (high grade), A (upper medium grade), BBB (lower medium grade), BB (non-investment grade speculative), B (highly speculative), CCC (substantial risk), CC (extremely speculative), SD (selective default), D (default) and NR (not rated). The + and − signs just reflect a little fine-tuning, so BB+ has a higher credit rating than BB, but not as good as BBB, etc.

(+) and (−) indicate positive (negative) outlook according to S&P.

Source: Standard & Poor's (2020).

other credit ratings in their home country. This means that a corporate borrower will most likely be downgraded following a downgrade of its sovereign country debt.

Table 11.1 shows sovereign credit ratings as of December 2020 according to Standard & Poor's, for a selection of countries. It can be seen that over the period studied China (A+) had a higher sovereign risk rating than India (BBB−, with a negative outlook), suggesting that in the credit-rating agency's view the Indian government is a higher credit risk than China's. The governments of Venezuela and Zambia, with credit ratings of SD, should be viewed as worse credit risks (i.e. of the countries listed in Table 11.1, these are the governments most likely to default on interest and principal repayment on their bond issues). A country rated SD (selective default) or D is in default on one or more of its financial obligations. As per May 2021, Lebanon and Suriname are also rated SD.

11.8 Operational and cyber risks

Another important risk in banking is **operational risk**. The Risk Management Group of the Basel Committee on Banking Supervision (2001a) defines operational risk as 'the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events'. In general terms, this is the risk associated with the possible failure of a bank's systems, controls or other management failure (including human error).

The definition of operational risk given above includes **technology risk**; however, there are some differences between the two that are outlined below (Saunders and Cornett, 2017):

- Technology risk occurs when technological investments do not produce the anticipated cost savings in the form of either economies of scale or scope; this risk also refers to the risk of current delivery systems becoming inefficient because of the developments of new delivery systems.
- Operational risk occurs whenever existing technology malfunctions or back-office support systems break down.

Operational risk was one of the main innovations in Basel II, requiring banks to hold capital for such risks along with credit and market risk; previously operational risk was covered by capital provisions set aside for credit and market risk. Today it includes a considerable portion of banks' risk-weighted assets, second only to credit risk (Aldasoro *et al*. 2020). Table 11.2 shows a list of operational risk event types that the Basel Committee, in consultation with the industry, has identified as having the potential to result in substantial losses.

As shown in the table, operational risk includes a number of risk event types, from employee fraud to natural disaster that may damage a bank's physical assets and reduce its ability to communicate with its customers.

In 2020, **cyber risks** (or IT disruption) were identified as the highest operational risk concerns, followed by data compromise, and theft and fraud (Risk.net, 2020). Like operational risk, cyber risk can involve the failure of some process or technology that could damage a firm. Kashyap and Wetherilt (2019) present many reasons why cyber risks are 'special' and should be appropriately regulated. Specifically, the authors warn how cyber risks differ from

Table 11.2 Operational risk event types

Risk event types	Examples include
Internal fraud	Intentional misreporting of positions, employee theft, and insider trading on an employee's own account.
External fraud	Robbery, forgery, cheque kiting* and damage from computer hacking.
Employment practices and workplace safety	Workers' compensation claims, violation of employee health and safety rules, organised labour activities, discrimination claims and general liability.
Clients, products and business practices	Fiduciary breaches, misuse of confidential customer information, improper trading activities on the bank's account, money laundering and sale of unauthorised products.
Damage to physical assets	Terrorism, vandalism, earthquakes, fires and floods.
Business disruption and system failures	Hardware and software failures, telecommunication problems and utility outages.
Execution, delivery and process management	Data entry errors, collateral management failures, incomplete legal documentation, unapproved access given to client accounts, non-client counterparty mis-performance and vendor disputes.

Note: * 'Cheque kiting' is a fraudulent method to draw against uncollected bank funds.

other operational risks both in the form they take and the potential vast impact they can have. The authors also report that financial services firms spend about 12 per cent of their IT budget on cyber security.

11.9 Off-balance-sheet risk

So far, we have analysed banking risks that arise through on-balance-sheet activities. For example, bad loans on the asset side, and deposit withdrawals and bank runs on the liability side. But they can also derive from off-balance-sheet exposures that can potentially translate into significant losses.

Off-balance-sheet (OBS) risk relates to the risks incurred when banks deal in activities that imply the increase in contracts that obligate them to perform in various ways but do not appear in the bank balance sheet. Examples of such activities are financial guarantees and letters of credit (as described in Chapter 10) that involve the creation of contingent assets and liabilities. It is also reasonable to think that the volatility in market values and complexity of many derivative instruments implies that they carry a considerable degree of OBS risk. This may still occur despite new accounting standards requiring banks to disclose the fair value of their financial derivatives in their balance sheet.

Given the nature of OBS activities and the fact that they have become increasingly widespread, it can be difficult for investors and regulators to identify the actual level of risk a bank is taking in any given period. Paradoxically, derivative products were introduced to reduce the amount of risk taken by a business, but recent history has shown that while they can provide some protection against important risks like interest rate or currency risk, excessive OBS commitments can result in spectacular losses, due to mismanagement or speculative use of financial instruments. The Barings Bank failure in 1995 is an often-quoted example of insolvency of a British investment bank due to the misuse of derivatives activities. The Barings case as well as some other well-known examples of losses caused by 'rogue traders' in derivatives and other OBS items (foreign exchange trading and investment fund business) are discussed in Chapter 8 (Box 8.2).

Given the potential huge losses that can be derived from excessive OBS trading, the regulators' approach has been to incorporate OBS commitments (using an on-balance-sheet or credit equivalent) in calculating bank capital adequacy requirements (see Chapter 7). In addition, as mentioned above, the most recent approach to accounting standards has been to require banks to disclose the amount of derivative commitments in the balance sheet at fair value.

11.10 Sustainability (or ESG) risks

Sustainability risks are defined as meaning 'an environmental, social or governance event or condition that, if it occurs, could cause a negative material impact on the value of the investment'.[3] At the bank level, sustainability risks are associated with environmental, social

[3] Definition by the European Union's SFDR (Sustainable Finance Disclosure Regulation), also known as Disclosure Regulation, which came into effect in March 2021. SFDR imposes new transparency disclosures rules and periodic reporting requirements on asset managers.

and governance (ESG) factors, which can have adverse effects on a bank's assets, both financial and intangible, such as reputation, as well as the bank's performance. There is also a potential for systemic risk from environmental and social factors, which could pose a threat to global financial stability.

ESG factors include (European Banking Authority, 2020):

- Environmental factors that refer to climate change mitigation and adaptation, as well as the environment more broadly and related risks (e.g. natural disasters);
- Social factors that refer to issues of inequality, inclusiveness, labour relations, investment in human capital and communities;
- Governance factors relate to the governance of public and private institutions and include management structures, employee relations and executive remuneration.

Environmental and social considerations are often interconnected. For example, climate change can increase existing inequalities; in addition, the governance factor has a central role in including social and environmental concerns in firms' decision-making process.

Today bank managers are under increased pressure to identify conditions and events that are linked to sustainability issues and pose risks that can materialise over the long term, such as climate change.[4] The European Banking Authority (EBA, 2020) distinguishes between two types of risk that act as main transmission channels for environmental risks:

- **physical risks**, which arise from the physical effect of climate change (e.g. extreme weather events and gradually deteriorating conditions in climate); and
- **transition risks**, which arise from the transition to a low-carbon and climate resilient economy (such as regulatory restrictions or taxation, disruptive technologies and changing consumer preferences).

Figure 11.5 shows theoretical examples of the impact of environmental factors through physical and transition risks (a and b) and social risks (c) on an institution's balance sheet (EBA, 2020).

Banks have a crucial role in financing sustainable growth because they are the biggest providers of capital for countries all over the world, so recognising the risks associated with ESG factors are very important in the transition to a more sustainable economy. For example, their decisions not to lend to, or invest in, businesses that are unsustainable (e.g. the

[4] Larry Fink, the founder, chairperson and CEO of BlackRock wrote in the 2019 annual letter to CEOs: 'As we enter 2019, commitment to a long-term approach is more important than ever—the global landscape is increasingly fragile and, as a result, susceptible to short-term behaviour by corporations and governments alike.'

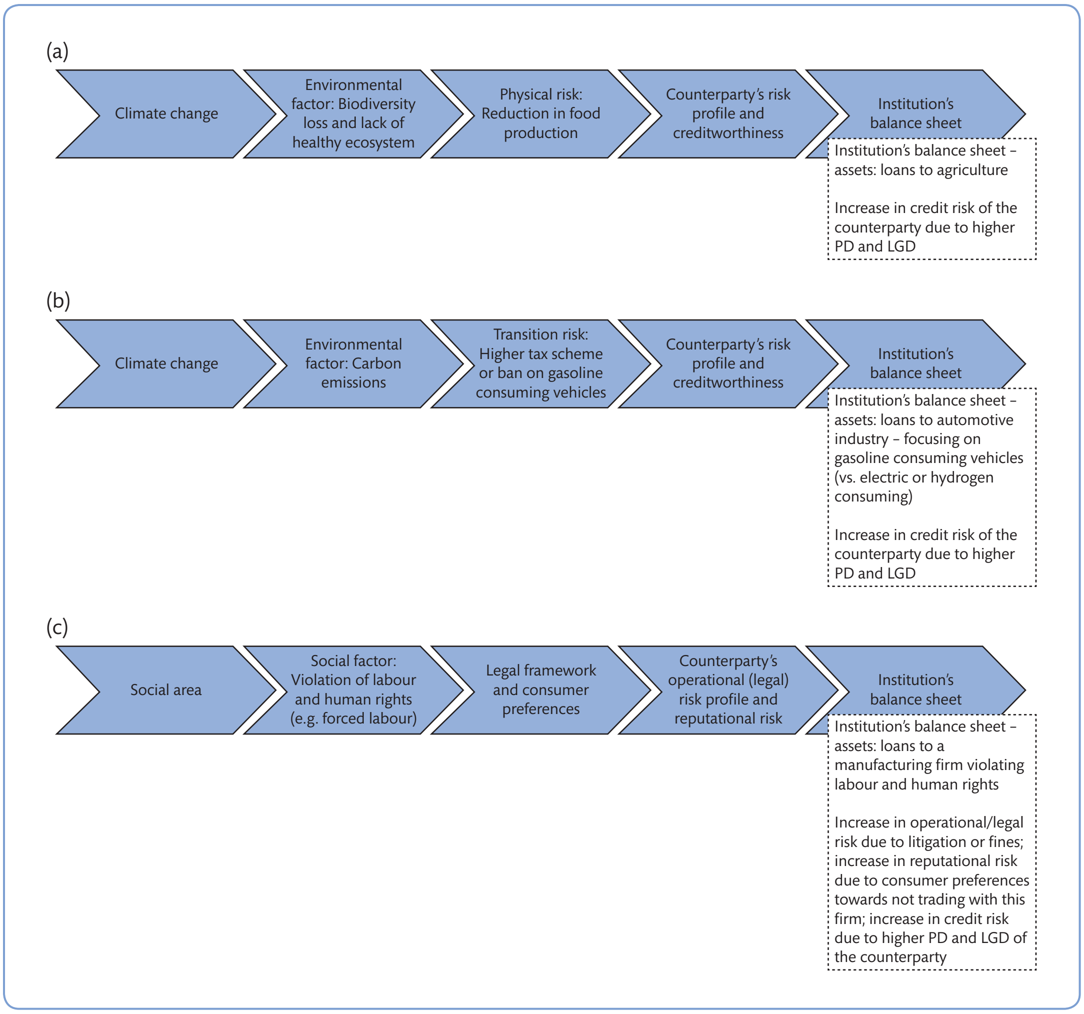

Figure 11.5 Theoretical examples on the ESG cycle – impact of environmental and social factors on an institution's risk profile: (a) impact through physical risk; (b) impact through transition risk; and (c) impact of social factors

Source: European Banking Authority (2020).

Note: PD= Probability of Default; LGD= Loss Given Default

fossil fuel industry) can support the switch to a low-carbon economy in line with the **Paris Agreement** (more details in Box 11.8). Figure 11.6 shows how **ESG risks** can be considered at the point of loan origination.

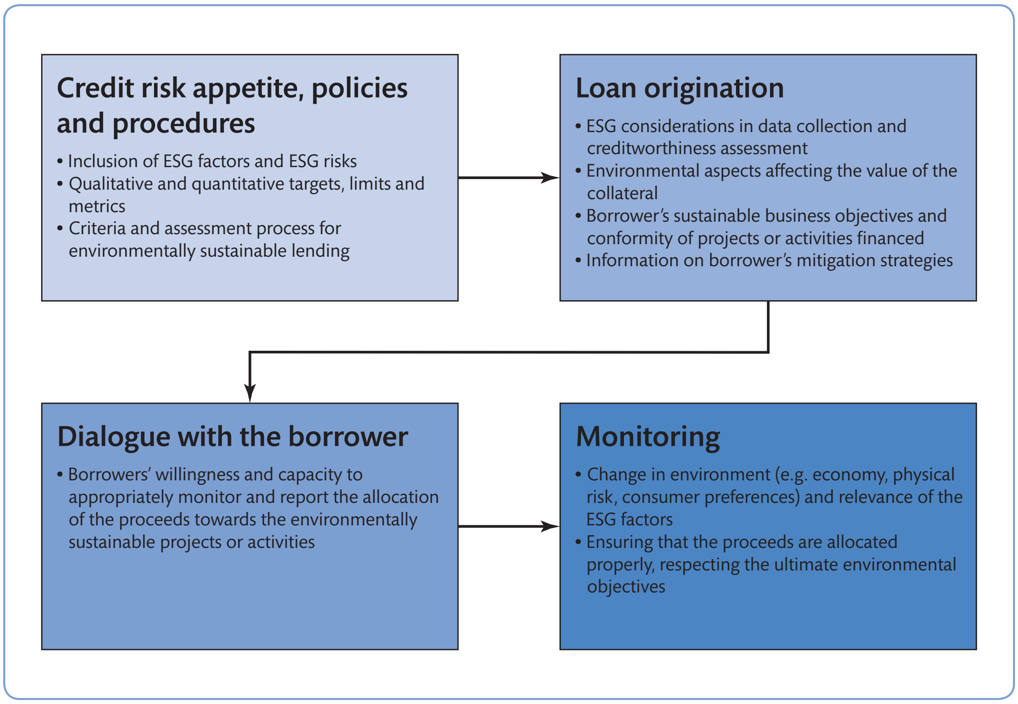

Figure 11.6 Incorporation of ESG risks at the point of loan origination

Source: European Banking Authority (2020).

BOX 11.8 THE VISION FOR SUSTAINABLE FINANCE IN THE EU

The Paris climate change agreement and the **United Nations 2030 Agenda for Sustainable Development**, which were finalised in 2015, represent universal visions and ambitious policy agendas to achieve the objectives of sustainability and climate change missions.

The United Nations **Sustainable Development Goals (SDGs)** list 17 targets as the blueprint for achieving a better and more sustainable future. These address challenges such as poverty, inequality, peace, justice and environmental issues.[5]

In line with these visions, since 2018 the European Commission has developed a comprehensive policy agenda on sustainable finance. Banks and other financial firms are expected to have a crucial role to play, specifically to:

- re-orient investments towards more sustainable technologies and businesses;
- finance growth in a sustainable manner over the long term;
- contribute to the creation of a low-carbon, climate-resilient and circular economy.

These principles are encapsulated in the 2018 Action Plan on Financing Sustainable Growth, which aims to reform the financial system in support of the EU's climate and sustainable development agenda (European Commission, 2018). The Action Plan is

[5] The United Nations SDGs are: (1) no poverty; (2) zero hunger; (3) good health and well-being; (4) quality education; (5) gender equality; (6) clean water and sanitation; (7) affordable and clean energy; (8) decent work and economic growth; (9) industry, innovation and infrastructure; (10) reduced inequalities; (11) sustainable cities and communities; (12) responsible consumption and production; (13) climate action; (14) life below water; (15) life on land; (16) peace, justice and strong institutions; and (17) partnerships (see **www.un.org/sustainabledevelopment/sustainable-development-goals/**).

BOX 11.8 The vision for sustainable finance in the EU (*continued*)

also linked to the Capital Markets Union (CMU)'s efforts to connect finance to the specific needs of the European economy to the benefit of the planet and our society. Figure 11.7 provides a visualisation of the actions towards the three objectives.

Sustainable finance and sustainable investment are also at the heart of the 2019 European Green Deal, which is a roadmap meant to foster the transition towards carbon neutrality by 2050.

The European Banking Authority (EBA) in December 2019 published an action plan specifically around sustainable finance to be completed by 2025. The plan outlines deliverables and activities on ESG factors and ESG risks. Its priorities include considerations on how ESG can be incorporated into the regulatory and supervisory framework of EU credit institutions. It also emphasises three areas where banks are encouraged to take actions: (1) strategy and risk management, (2) disclosure and (3) scenario analysis.

In the framework of the European Green Deal, the Commission also published a Renewed Sustainable Finance Strategy and implementation of the action plan (August 2020), which aims to provide the policy tools to ensure that the financial system supports the transition of businesses towards sustainability during the period of recovery from the impact of the Covid-19 pandemic.

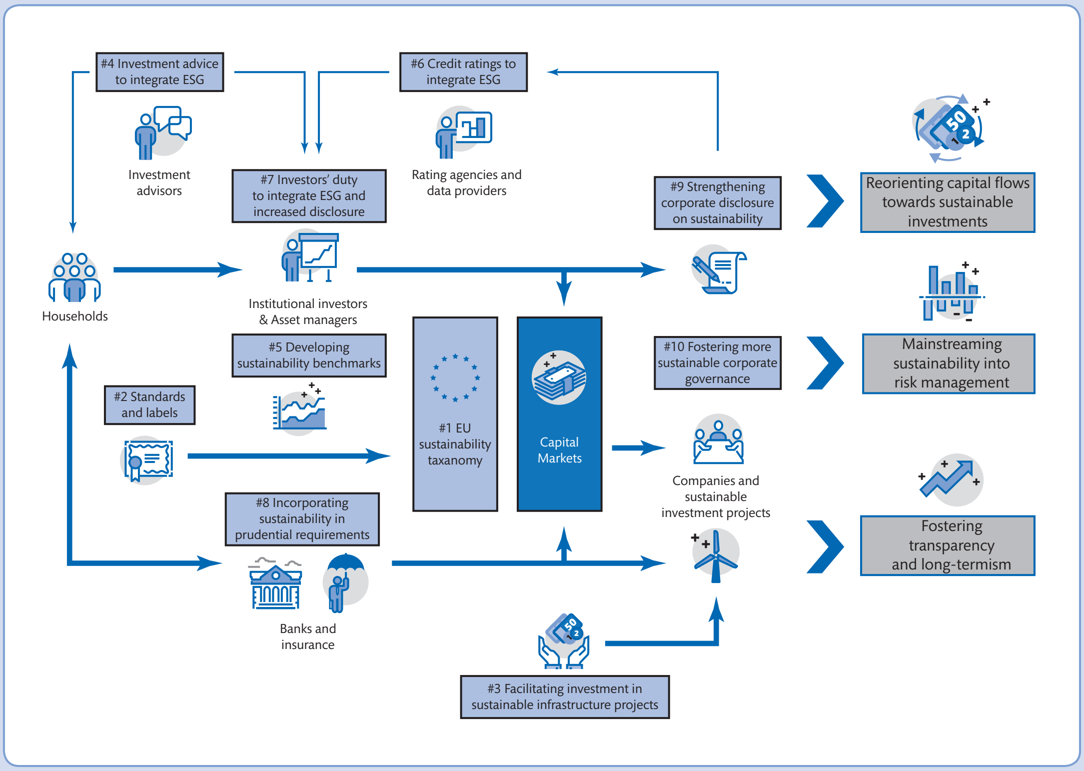

Figure 11.7 Visualisation of the 2018 EU action plan on financing sustainable growth

Source: European Commission (2018).

11.11 Other risks

There are a number of other risks that banks face. It is common to distinguish between macro- and micro-risks. Macro-risks are also known as environmental risks and some of them are common to all businesses. They can include the risk of an economic recession, a sudden change in taxation or an unexpected change in financial market conditions, due for example to war, revolution, stock market crashes or other factors. Additional examples include:

- **Inflation risk** – the probability that an increase in the price level for goods and services will unexpectedly erode the purchasing power of a bank's earnings and returns to shareholders.
- **Settlement or payment risk** – a risk typically faced in the interbank market; it refers to the situation where one party to a contract fails to pay money or deliver assets with another party at the time of settlement. It can include credit/default risk if one party fails to settle. It can also be associated with any timing differences in settlement between the two parties.
- **Regulatory risk** – the risk associated with a change in regulatory policy. For example, banks may be subject to certain new risks if deregulation takes place and barriers to lending or to entry of new firms are lifted. Changing rules relating to products and dealing with customers are other examples of potential regulatory risk.
- **Competitive risk** – the risk that arises as a consequence of changes in the competitive environment, as bank products and services become available from an increasing number of new entrants including non-bank financial firms and retailers.

Microeconomic risks are generally due to factors inside the bank, rather than external factors, such as:

- **Operating risk** – the possibility that operating expenses might vary significantly from what is expected, producing a decrease in income and the bank's value.
- **Legal risk** – the risk that contracts that are not legally enforceable or documented correctly could disrupt or negatively affect the operations, profitability or solvency of the bank.
- **Reputation risk** – the risk that strategic management's mistakes may have consequences for the reputation of a bank. It is also the risk that negative publicity, either true or untrue, adversely affects a bank's customer base or brings forth costly litigation thereby affecting profitability.
- **Portfolio risk** – the risk that the initial choice of lending opportunities will prove to be poor, in that some of the alternative opportunities that were rejected will turn out to yield higher returns than those selected.
- **Call risk** – the risk that arises when a borrower has the right to pay off a loan early, thus reducing the lender's expected rate of return.

Sometimes risks may be due to both internal and exogenous factors. *Earnings risk*, for example, is the risk that earnings may decline unexpectedly and this may be due to bad management or a change in laws and regulation.

Finally, it is worth mentioning **management risk**. This is the risk that management lacks the ability to make commercially profitable and other decisions consistently. It can also include the risk of dishonesty by employees and the risk that the bank will not have an effective organisation.

11.12 Capital risk and solvency

Capital risk should not be treated as a separate risk because all risks described in this chapter can potentially affect a bank's capital. In other words, excessive credit risk, interest rate risk, operational risk, liquidity risk, OBS risk, etc. could all result in a bank having insufficient capital to cover such losses. As noted in Chapter 8 (Box 8.2) in the case of Barings Bank, in such an instance the bank's solvency is impaired and failure can occur.

A bank will be insolvent when it has negative net worth of stockholders' equity. We learnt in Chapter 9 that this is the difference between the market value of its assets and liabilities. Therefore, capital risk refers to the decrease in market value of assets below the market value of its liabilities. In case of liquidation the bank would not be able to pay all its creditors and would be bankrupt.

Capital risk is closely tied to financial leverage (debt/equity) and banks are typically highly leveraged firms. It also depends on the asset quality and the overall risk profile of the institution. One should realise by now that the amount of capital a bank holds is positively related to the level of risk, that is, the more risk taken, the greater the amount of capital required. Banks with high capital risk (that is, banks that have low capital to assets ratios) also normally experience greater periodic fluctuations in earnings.

Capital risk, therefore, is the same as the risk of insolvency or the risk of failure. The following are some early indicators of failure risk (Rose and Hudgins, 2010, p. 187):

- The interest rate spread between market yields on bank debt issues and market yields on government securities of the same maturity: if the spread increases then investors believe that the bank in question is becoming more risky relative to government debt – investors in the market expect a higher risk of loss from purchasing and holding the bank's debt.
- The ratio of stock price per share to annual earnings per share: the ratio will often fall if investors come to believe that a bank is undercapitalised relative to the risks it has taken on.
- The ratio of equity capital to total assets: a low level of equity relative to assets may indicate higher risk exposure for debt holders and shareholders.
- The ratio of purchased funds to total liabilities: purchased funds usually include uninsured deposits and short-term borrowings in the money market.
- The equity capital to risk-weighted assets ratio: indicates how well a bank's capital covers potential losses from the assets that are most likely to decline in value.

11.13 Interrelation of risks

In this chapter we have presented the characteristics of the main risks that modern banks face. While we have described each risk independently, risks are often correlated. For example, an increase in interest rates may increase default risk because companies may find it more difficult to maintain the promised payments on their debt. It may also increase liquidity risk for the bank if the defaulted payments were important for liquidity management purposes. Ultimately, this will affect bank's earnings and capital. Similarly, environmental, social and governance risks are closely interrelated. Box 11.9 provides some examples of the social impact of Covid-19.

BOX 11.9 EXAMPLES OF THE SOCIAL IMPACTS OF THE COVID-19 PANDEMIC

The outbreak of the Covid-19 pandemic provides a good example of the interaction between environmental, social and governance factors. Several commentators have highlighted the importance of biodiversity loss in the origin and spread of new diseases with several health and social impacts (European Commission, 2020).

From a social perspective, the widespread containment measures introduced to limit the spread of the disease have severely impacted our way of life. In addition to the profound impact of the economic disruption and the associated unemployment, the term 'social distancing' suddenly became well known to billions of people, who were asked to stay at home, refrain from social contacts and observe strict rules for non-avoidable social interaction. Shortly after the outbreak, several studies have highlighted the social consequences of such containment measures.

The financial impact of the pandemic has already been visible on the balance sheets of institutions in the first quarter results of 2020. The risk driver of the impact has been widely associated with the increased credit risk of the counterparties, which suffered from the suppression of economic activity during the confinement and higher levels of unemployment. However, even after the lifting of the confinement measures, several companies continued to suffer from turnover below average, pointing to the role of social dynamics on economic behaviours. While a precise quantification of the financial impact of the social dynamics is still difficult, there are some areas where the impact for institutions of social distancing measures can be better assessed.

The sudden introduction of social distancing measures forced companies and businesses to either adjust the organisation of their workforce and processes or shut down. Sometimes overnight, a significant proportion of people started smart working, with also the most reluctant companies being forced to adjust. Months of mandatory 'work-from-home' provided a natural experiment to assess not only the impacts on productivity, but also on well-being, work–life balance and diversity. A number of companies have announced long-term plans to shift a significant share of their employees to long-term smart working. Another side effect of the Covid-19 pandemic is the necessary reorganisation of work processes to comply with distancing measures.

Source: Adapted from European Banking Authority (2020).

Note that banks face a plethora of risks and these risks are not mutually exclusive. They face credit risk when lending, market risk when trading in securities, interest rate risk in their funding operations, operational risks in undertaking all activities, and so on.

11.14 Conclusion

Risks are part of any economic activity. In financial services they assume a particular relevance because the business of banking is linked to uncertain future events and the risk of a bank failure is always a possibility in a system essentially run on confidence.

We have shown in this chapter that banks have to face a number of often interrelated risks, such as interest rate risk, credit risk and liquidity risk and that banks' attempts to cope with risks have often brought about other risks. One such case is derivative products that were designed to control risk through various hedging techniques, and have demonstrated, in certain cases, to be highly risky instruments. In this chapter we have also highlighted how banks today need to learn to recognise the risks associated with ESG factors in the transition to a more sustainable economy.

There are various ways for a well-managed bank to protect against risk. One way is through diversification, in all its forms (e.g. portfolio diversification and geographic diversification). In addition, a bank will have to carry out appropriate asset/liability management practices and hedging strategies. Finally, there is capital that can act as a financial buffer, thus minimising the likelihood of bank failure. The next chapter discusses these and other issues and focuses on the various approaches to managing the main risks in modern banking.

Key terms

Banking book
Call risk
Capital risk
Competitive risk
Counterparty risk
Country risk
Credit culture
Credit philosophy
Credit-rating agencies
Credit risk
Cyber risk
ESG risks
Fixed-rate assets and liabilities
Foreign exchange risk
Inflation risk
Interest rate risk
Legal risk
Liquidity crisis
Liquidity risk
Loan policy
Management risk
Market risk
Off-balance-sheet (OBS) risk
Operating risk
Operational risk
Paris Agreement
Physical risk
Portfolio risk
Rate-sensitive assets and liabilities
Refinancing risk
Regulatory risk
Reinvestment risk
Reputation risk
Settlement or payment risk
Sovereign risk
Sustainability risks
Sustainable Development Goals (SDGs)
Systematic/unsystematic credit risk
Technology risk
Trading book
Transition risk
United Nations 2030 Agenda for Sustainable Development

Key reading

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Saunders, A. and Cornett, M.M. (2017) *Financial Institutions Management: A Risk Management Approach*, New York: McGraw Hill.

REVISION QUESTIONS AND PROBLEMS

11.1 What is credit risk and what is meant by credit culture?

11.2 Define interest rate risk and distinguish between reinvestment and refinancing risk.

11.3 What are rate-sensitive assets and liabilities?

11.4 Why is liquidity risk one of the most important concerns for bank management?

11.5 Is it correct to state that the root of liquidity risk lies on information asymmetries?

11.6 What is exchange rate risk? What are the main features of foreign exchange markets?

11.7 What is a commercial bank's banking book? Why should the boundaries between banking books and trading books be clearer?

11.8 What are the differences between country and sovereign risk?

11.9 What distinguishes operational risk from technology risk? What are cyber risks?

11.10 Explain the importance of capital risk.

11.11 What are sustainability risks? Can you give examples of ESG factors?

11.12 Read the articles 'Barclays pressed to stop financing some fossil fuel groups' (*Financial Times*, 8 January 2020) and 'HSBC targeted by shareholders over fossil fuel financing' (*Financial Times*, 10 January 2021). Discuss the role of banks in financing sustainable investments.